CSE 317
Homework 1

June 18, 2025

Dear students, There are a total of 6 questions in this homework. We invalidate the grouping; instead, it will be an individual assignment where each student has to submit individually. Each student must solve the $(ddd\%6+1)$th problem and one other problem of their choice. Here, "ddd" is the number formed by the last three digits of the ID. Thus, a student with ID 2105001 will solve $(1\%6+1)$ = the 2nd problem and one problem from the other 5 problems by his own choice. Thus, everyone must solve two problems. Only a handwritten solution will be accepted. Collaboration between two students is strictly prohibited. Upload the scanned final PDF here. Identical or highly similar solutions will result in negative marks. Please consider that these homework assignments will have significant weight in your final grading. Please don't ask AI to solve your exercises. We don't expect perfect solutions. An incomplete/imperfect solution with real effort may result in full marks, while a complete, perfect AI-generated solution will result in negative marks. Moreover, similar questions will arrive in the CTs and the final exam. Your submission file should be like "2105ddd-HW1.pdf" for the student with ID 2105ddd.

1. Q1: How many possible decision trees to represent the hypothesis space of all Boolean functions with n input variables? You need to try to give a lower bound and an upper bound on the number of such decision trees possible. Your answer should be rigorously mathematically sound.

2. Q2: Explain the χ^2 pruning technique of decision tree, deriving the measure of deviation? Why should we not combine this pruning technique with information gain to perform early stopping?

3. Q3: Variance means that a learning algorithm will in general, return different hypotheses for different sets of examples. Prove that if the problem is realizable, then the variance decreases towards zero as the number of training examples increases. You may consider any necessary assumption. Your proof should be mathematically sound.

4. Q4: Prove that finding a minimal decision tree consistent with a set of data is NP-hard.

5. Q5: Theoretically prove that the Gini index and majority error are approximations of entropy. Describe the scenarios in which the approximation is good. Could you describe any scenario where a majority error-based ID3 algorithm will return a different tree than the entropy-based algorithm?

6. Q6: A decision list consists of a series of tests, each of which is a conjunction of literals. If a test succeeds when applied to an example description, the decision list specifies the value to be returned. If the test fails, processing continues with the next test in the list. Decision lists resemble decision trees, but their overall structure is simpler: they branch only in one direction. In contrast, the individual tests are more complex.

   (a) If we allow tests of arbitrary size, then prove that decision lists can represent any Boolean function.

   (b) We use the notation k-DL for a decision list with up to k conjunctions. We use the notation k-DL(n) to denote a k-DL using n Boolean attributes. Show that k-DL includes as a subset k-DT, the set of all decision trees of depth at most k.

   (c) Show that the number of examples needed for PAC-learning a k-DL(n) function is polynomial in n.

Deadline: 5th July 11:59 PM